Surendra Bhattarai

Astronomy, Yale University | surendra.bhattarai@yale.edu | surendra-bhattarai.github.io | LinkedIn | Google Scholar

Education

Yale University, Ph.D. in Astronomy (ongoing), advised by Prof. Sarbani Basu — 2024 –

Yale University, MS and M.Phil in Astronomy (expected 2026) — 2024 – 2026

Indian Institute of Science, Bangalore, Master's thesis in Physics, advised by Prof. Banibrata Mukhopadhyay — 2023 – 2024

Indian Institute of Science Education and Research (IISER) Kolkata, BS-MS in Physics — 2019 – 2024

First-author Publications

h-index: 6 | Google Scholar

2. **Bhattarai S**, Basu S, *Empirical correction for $\nu_{\max}$ scaling relation* (in prep.)

1. **Bhattarai S**, Bellinger E P, Farag E, Bauer E B, et al., *Asteroseismology of polluted white dwarfs* (submitted to *ApJ*)

Other Publications

4. Zuraiq Z, Kumar A, Hackett A J, **Bhattarai S**, Tout C A, Mukhopadhyay B, *Simulating super-Chandrasekhar white dwarfs*, in The Relativistic Universe: From Classical to Quantum (ISRA 2023), Astrophysics and Space Science Proceedings, vol. 61, pp. 397–407, Springer Nature (2025)

3. Mistry P, Prasad A, Maity M, ..., **Bhattarai S**, et al., *VaTEST III: Validation of 8 potential super-Earths from TESS data*, PASA, 41, e030 (2024)

2. Mistry P, Pathak K, Prasad A, Lekkas G, **Bhattarai S**, et al., *VaTEST II: Statistical Validation of 11 TESS-detected exoplanets orbiting K-type stars*, AJ, 166, 9 (2023)

1. Mistry P, Pathak K, Lekkas G, Prasad A, **Bhattarai S**, et al., *VaTEST I: Validation of sub-Saturn exoplanet TOI-181b in narrow orbit from its host star*, MNRAS, 521, 1066 (2023)

Fellowships and Awards

McDougal STEM Career Fellowship, Office of Career Strategy, Yale University ($6,490) — 2026

Boris Garfinkel Prize Fellowship in Astronomy, Yale University ($2,500) — 2024

Yale Graduate School Fellowship, Yale GSAS — 2024 –

INSPIRE-DST Fellowship, Government of India (₹400,000) — 2019 – 2024

Anundoram Borooah Award, Government of Assam — 2017

Research Experience

Graduate Researcher, Yale University — 2024 –

- Asteroseismic ages of open clusters from TESS and Kepler red giants
- Tested the $\nu_{\max}$ scaling relation and provided a correction; asteroseismology of polluted white dwarfs

Master's Thesis Researcher, Indian Institute of Science, Bangalore — 2023 – 2024

- Super-Chandrasekhar white dwarfs with magnetic fields and rotation, modeled in `MESA` and `STARS`

Visiting Researcher, ICRAR, University of Western Australia, with L. Cortese — 07/2023 – 12/2023

Undergraduate Researcher, IISER Kolkata, with S. K. Garain — 01/2023 – 05/2023

Summer Researcher, IUCAA Pune, with A. Ganguly — 07/2022 – 10/2022

IAstro Summer Intern, Instituto de Astrofísica e Ciências do Espaço, Lisbon — 07/2022 – 09/2022

Co-lead, Project VaTEST, student-led TESS exoplanet validation — 2022 – 2024

Professional Experience and Service

McDougal STEM Career Fellow, Office of Career Strategy, Yale University — 2026 –

Senator, Yale Graduate and Professional Student Senate (GPSS) 2026 –

Member, Astronomy Climate and Diversity Committee, Yale Astronomy 2025 –

Local Organizing Committee, 42nd Meeting of the Astronomical Society of India, IISc 2024

Subject Matter Expert (Mathematics), Chegg 2022 – 2024

Talks

Asteroseismology of polluted white dwarfs, Nuclear Asteroseismology Workshop, Yale 07/2026

Can eccentric GW signals mimic microlensed signals?, IUCAA Pune (video) 08/2022

Passive, but not resigned: galaxies moving from star-forming to quiescent, IA Lisbon 07/2022

Teaching

Teaching Fellow, Yale University 2024 –

- *Stars and Their Evolution* (ASTR 2100), Fall 2026
- *Relativity and Black Holes* (ASTR 1800), Fall 2025
- *Galaxies and Cosmology* (ASTR 220), Spring 2025
- *Introduction to Astronomical Observing* (ASTR 155), Fall 2024

Instructor (Python and observing), Leitner Observatory and Planetarium, Yale 2024 –

Instructor, Astro Research Preparation Series, Yale University Spring 2025

Observing Mentor, Yale Summer Program in Astrophysics (YSPA) Summer 2025

Teaching Assistant, *Computational Physics* (PH3205), IISER Kolkata Spring 2023

Outreach

Space Policy Advocate, Yale STEM Hill Day, Washington, DC 03/2025

Volunteer, IISc Open Day, Bangalore; National Space Science Exhibition, Kolkata 2022 – 2024

Volunteer Tutor, Ek Pehal (education for underprivileged children), Kolkata 2019 – 2022

Selected Schools and Workshops

MESA Summer School, University of Wyoming 06/2026

Tinsley Workshop, Yale University 10/2024

Sagan Exoplanet Summer Workshop, NExScI 07/2023

Introductory Summer School in Astronomy and Astrophysics, IUCAA Pune 06/2022

SPARC Workshop on Machine Learning in Solar Physics, IISER Kolkata 06/2022

Radio Astronomy Winter School, NCRA Pune 12/2021

Skills

Statistics and ML: Bayesian inference, MCMC, nested sampling, regression, machine learning (gradient boosting, Shapley values), time-series and Fourier analysis, uncertainty quantification

Programming: Python (NumPy, SciPy, pandas, scikit-learn, Matplotlib, Astropy, lightkurve), Fortran, C, MATLAB, Bash, Git, Linux, SLURM/HPC

Scientific codes: `MESA` (stellar evolution), `GYRE` (stellar oscillations), `STARS`, `PBjam`

References

Prof. Sarbani Basu, Department of Astronomy, Yale University

Prof. Earl Bellinger, Department of Astronomy, Yale University

Prof. Banibrata Mukhopadhyay, Department of Physics, Indian Institute of Science Bangalore